

VisionGuard

Catching vehicle theft & break-ins the moment they happen — on the cheap, dark cameras societies already own.

Concept note & prototype status · Founder: [your name] · [email / phone] · [FILL: date]

The problem

The world has close to a billion CCTV cameras, and almost all of them are **passive** — they record for review *after* a crime, not prevention *during* one. In residential societies, dozens of cameras run that nobody watches. When a car is broken into at night, the guard finds out the next morning, from the footage. The camera saw everything and did nothing.

Existing "AI cameras" don't fix this because they are built and demoed on clean daylight video. Real CCTV is **low-resolution, dark, and low-frame-rate** — exactly where they fail. And generic motion alerts are so noisy that people switch them off. A noisy security system is worse than none.

The solution

VisionGuard turns a society's existing cameras into a watchful AI that:

- **Sees in the dark** — enhances low-light frames so it detects the person a raw feed hides.
- **Understands the act** — fuses a person at a vehicle, a localized break-in motion, and the car leaving into a single *theft* conclusion.
- **Explains itself** — Claude reviews the flagged clip and writes a plain-English verdict with timestamps and severity.
- **Alerts instantly** — owner and guard are notified at the smash, not the next morning.

How it works

Local-first: cheap detection runs on-site; only flagged evidence goes to Claude for a verdict; **raw CCTV never leaves the box**.

VisionGuard — how it works

Local-first AI security. Cheap detection runs on-site; Claude reviews only the flagged moments; raw CCTV never leaves the box.

ON-PREMISE · RUNS ON-SITE, NO CLOUD DEPENDENCY

1 · SEE

CCTV cameras

Any RTSP / IP camera. **Low-res, dark, low-fps** — the real-world quality we tuned for.



Low-light enhancement

CLAHE + gamma reveal **dark, small subjects** a raw frame hides. Proven on real 452x342 night footage.



Detect & track

YOLO11 + ByteTrack give every person & vehicle a **persistent ID** and trajectory.

2 · UNDERSTAND (THE FREE LAYER + HYBRID BRAIN)

Free layer

Pose, proximity, dwell, zones, and a **localized break-in detector** (peak motion at the window) — cheap signals, per-camera state.



Specialist models

R3D-18 CNNs score 4-sec clips for **break-in** & **vehicle tampering**, with per-camera temporal confirmation.



Evidence fusion

No single signal can alarm. Fuses model + motion + relationship + history → **NORMAL · WATCH · REVIEW · CONFIRMED**.



Incident memory

Collapses a whole theft into **one explainable incident** — not fifty alerts.

Only evidence leaves the box

— selected keyframes + metadata, never continuous footage. →

SECOND STAGE · PAY ONLY FOR FLAGGED MOMENTS

Claude verdict

Reads the flagged clip and writes a **plain-English verdict**: “car driven away after tampering — likely theft,” with timestamps & severity.



Operator dashboard

Incident cards, annotated video, jump-to-moment — for guards, managers, admins.



Instant alerts

Telegram now; email / SMS / app via webhooks. Owner & guard notified at the smash, not after.

Cost model: detection is free (on-site). Claude review runs only on flagged clips — paise per incident, not a per-camera cloud subscription.

The VisionGuard pipeline: enhance → detect & track → free-layer signals + specialist models → evidence fusion → incident memory → Claude verdict → dashboard & alerts.

It works on real, bad footage — the evidence

We diagnosed and fixed detection on **actual 452×342 night-CCTV theft footage** (not a lab clip). Two measured results from that real video:



Same frame, before and after our low-light enhancement. In the raw feed the intruder at the car window is black-on-black and invisible to the detector; enhanced, the person and window are clearly visible.

101 → 142

FRAME BRIGHTNESS AFTER
ENHANCEMENT (PERSON
BECOMES DETECTABLE)

90–160

LOCALIZED BREAK-IN SIGNAL
DURING THE ACT (VS ~15
WHEN QUIET)

180

AUTOMATED TESTS PASSING
ACROSS THE PIPELINE

The break-in leaves a clean, well-separated signal at this exact quality — but only when measured as **localized** motion at the contact point, which we built. The whole system then collapses the event into one explainable incident and Claude writes the verdict.

What an operator sees

Forensic Lab

Upload CCTV footage — the free layer flags suspicious moments, and Smart AI Review makes Claude actually watch the video.



CLAUDE VERDICT: HIGH AT 60S — CAR DRIVEN AWAY BY THE PERSON WHO REACHED INTO IT — LIKELY THEFT

3 finding(s) — timestamps listed below

INCIDENTS DETECTED

INCIDENT 1 HIGH

POSSIBLE VEHICLE THEFT:
vehicle drove away 48.0s after
suspicious activity around it

🕒 11.0–60.0s 👤 culprit #7 ▶ 4 alert(s)

INCIDENT 2 MEDIUM

Suspicious activity: person at
vehicle window, localized
disturbance at the car

🕒 11.0–13.5s 👤 culprit #7 ▶ 2 alert(s)

The analysis view: Claude's plain-English verdict on top, then the theft as a single incident card (time span, culprit ID, severity) — not a wall of alerts. Click a card to jump the video to that moment.

Where the prototype is in the build

VisionGuard is a **working, end-to-end prototype**, validated qualitatively on real footage. It is honest about what is proven and what the next stage delivers.

AREA	STATUS	DETAIL
Detection on real low-res night CCTV	Built ✓	Low-light enhancement + YOLO/ByteTrack tracking; person & vehicle detection proven on the actual clip.
Break-in / theft understanding	Built ✓	Localized motion detector + free-layer signals + theft-chain, measured on real footage.
Specialist models + fusion	Built ✓	R3D-18 CNNs (break-in, vehicle tampering) with per-camera temporal confirmation; evidence fusion so no single signal can raise a false alarm.
Claude verdict & incident memory	Built ✓	Plain-English verdict per incident; per-frame alerts merged into one incident.
Operator dashboard + Telegram alerts	Built ✓	Upload-and-analyze "Forensic Lab", incident cards, annotated playback, instant alerts.
Published accuracy number	In progress ⌚	Requires an untouched, labeled multi-camera holdout — the deliberate gap this raise closes.
Live multi-camera & mobile app	Next →	Live pipeline, roles/permissions, mobile alerts, email/SMS routing.
Model hardening on domain data	Next →	Train on our own society footage + surveillance datasets (SPHAR, VIRAT) to cut domain-shift.

Tech: Python · FastAPI · YOLO11 + ByteTrack · R3D-18 (PyTorch) · OpenCV · SQLite · Claude (Anthropic) for scene-verdict review · Telegram alerts. Local-first architecture; footage stays on-site.

Why now, and where we win

Why now: cameras are everywhere and AI is finally good enough to watch them live — but nobody has made it work on the cheap, dark cameras people actually own. We have. **The market** (AI video surveillance) is already several billion dollars and growing fast.

Our white space vs. well-funded incumbents (Verkada, Ambient.ai globally; Staqu, Wobot, Awiros in India): most target enterprise/retail/city or sell their own expensive hardware. We serve **residential societies on their existing cheap cameras**, tuned for **low-quality footage**, with **explainable, local-first** alerts at a **paaise-per-incident** cost. The moat is our performance on bad footage plus the data flywheel from real deployments.

The ask

Raising [FILL: ₹ amount] to reach [FILL: milestone — e.g. validated recall + first paying societies]. Use of funds: data collection + GPU to harden the models, pilot deployments, and the product build (live multi-

camera, mobile, front-end).

VisionGuard — concept note & prototype status. Figures marked as measured are from testing on real low-resolution night CCTV and are engineering evidence, not a certified production-accuracy claim. Working prototype available for a live walkthrough on request.